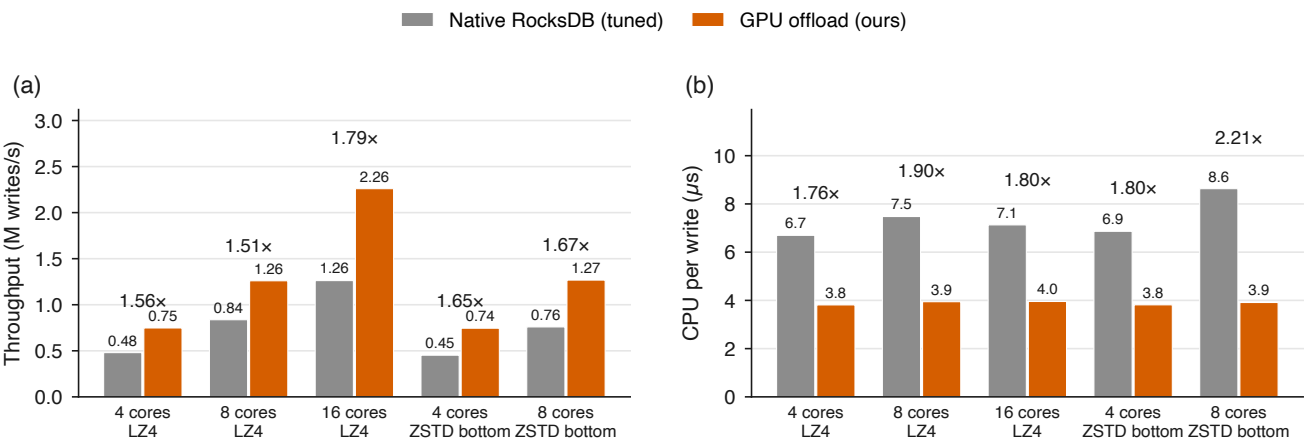
Native RocksDB (tuned)
GPU offload (ours)
(a)
Throughput (M writes/s)
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0.48
0.75
1.56×
0.84
1.26
1.51×
1.26
2.26
1.79×
0.45
0.74
1.65×
0.76
1.27
1.67×
4 cores LZ4
8 cores LZ4
16 cores LZ4
4 cores ZSTD bottom
8 cores ZSTD bottom
(b)
CPU per write (μs)
10
8
6
4
2
0
6.7
3.8
1.76×
7.5
3.9
1.90×
7.1
4.0
1.80×
6.9
3.8
1.80×
8.6
3.9
2.21×
4 cores LZ4
8 cores LZ4
16 cores LZ4
4 cores ZSTD bottom
8 cores ZSTD bottom